University: The University of Hong Kong (HKU)
Unit of Assessment (UoA): 13 – Computer Science / Information Technology
Title of case study: UTEE (Ubiquitous Trusted Execution Environments): A Secure, High-performance Distributed Bigdata Computing System on Public Clouds
Is this case study continued from a case study submitted in RAE 2020? N

1. Summary of the impact (indicative maximum 100 words)

Prof. Heming Cui invented UTEE, a novel software system for ensuring the confidentiality of bigdata computing tasks on public clouds, against vulnerabilities and bugs in the widely used Linux-based operating system kernel (consisting of 40 million of lines of code written by over 1500 companies and organizations in the world). His UTEE project made HKU receive in total HKD $7.5 million licensing fee from Huawei **[4.1]**, and he transferred 15 patents to Huawei. In Apr 2021, UTEE is commercialized as the essential runtime system of Huawei's major secure cloud service TICS (www.huaweicloud.com/product/tics.html); UTEE is invoked in every bigdata computing task of TICS **[3.2, 3.3]**. According to Huawei's official letter in 2021 (attached), UTEE has 3 million active cloud tenants (including independent companies and individuals) located in 170 countries **[5.8]**. Consider just one publicly known example about Huawei's many global strategic company partners: Huawei integrated UTEE (TICS) in the Huawei Cards released by the China Citic Bank in 2021 **[5.5]**, in order to protect users' data security on every credit card transaction. The number of users of this specific bank's Huawei Cards exceeded 2 millions in Sep 2023 **[5.6]**; the total number of Huawei Cards users exceeded 10 millions in Sep 2024 **[5.7]**. When every user invokes UTEE, compared to UTEE's most advanced competitors, UTEE greatly reduced the lines of software code that must be trusted from about 1050K to 250K (a 79% reduction) **[3.2, 3.3]**. Also, on the newest Intel CPUs, UTEE supports to execute three orders of magnitude larger dataset sizes, compared to the dataset sizes that are supported to execute by all the competitors **[3.2, 3.3]**.

2. Underpinning research (indicative maximum 500 words)

[Research background of TEE] In this bigdata era, people usually conduct computations on sensitive user data (e.g., decrypted data) using bigdata computing software (e.g., Spark and Flink), and the computation tasks of each bigdata software are spitted and ran on a massive number of (heterogeneous) CPUs and GPUs in a public cloud (e.g., Amazon Cloud, Huawei Cloud, and Hospitals that conduct big-data computations). It is well known that over 80% of computers on public clouds (including Amazon, Google, Huawei, and Alibaba) use Linux-based operating system (OS) kernel to run at the highest privilege level (including managing computing tasks); Linux kernel consists of 40 million of lines of code written by over 1500 companies and organizations in the world. Unfortunately, these huge lines of OS code imply that the OS is plagued with vulnerabilities and bugs that can easily leak user data to the Internet (the notorious "data breach" problem). According to IBM's 2024 Cost of a Data Breach Report, the world's financial loss incurred by every data breach is about USD $4.88 million. And ironically, the notorious Minnesota University (just one organization among the 1500) Linux Hypocrite Commits incident in Apr 2021 caused the injections and fixes of over 190 known vulnerabilities in the Linux kernel (maybe many unknown ones, injected by this incident, are still left in Linux).

To eliminate the OS kernel from the trusted computing base (i.e., from the risk of leaking user data on public clouds), in recent years, a powerful and hardware-enforced security technique called **TEE (Trusted Execution Environments)**, is available in high-end commercial CPU processors (e.g., Intel SGX and Arm TrustZone). TEE is a processor's special, tiny `execution zone', which is physically isolated from the rest of the processor. When a bigdata task runs in a TEE, TEE's hardware protects the confidentiality and integrity of the task and data, strictly against the entire cloud software and people who have high software privileges, including malicious/flawed software (e.g., OS kernels with bugs) and malicious cloud administrators.

Unfortunately, despite efforts of academic and industrial systems (including Microsoft's VC3, European Union's SGX-Spark, UC Berkeley's Opaque, Google's Confidential Computing, and UNC's Civet), all existing TEE-relevant systems still face three significant open research problems: (1) confirmed by all these competitor systems, TEE is still a **low-level** (assembly or C languages) programming object, so it is extremely hard to use by regular cloud software developers; (2) confirmed by our [Cronous ACM Micro 2022] paper in Section 3, TEE is only commercially available in some CPUs, but **not in general GPUs**; (3) revealed by our [Uranus ACM AsiaCCS 2020] paper in Section 3, existing competitors' TEE systems suffer from **20X~100X performance overhead** compared to the native (insecure) bigdata computations, and **cannot support typical big-data datasets** (e.g., terabyte).

According to our published research paper and patent **[3.2, 3.3]**, UTEE greatly reduced the trusted computing base of a system invoking it. Specifically, although both UTEE and its worldwide most advanced competitors can eliminate the Linux kernel from the trusted computing base,; When every user invokes UTEE, compared to UTEE's most advanced competitors, UTEE greatly reduced the lines of software code that must be trusted from about 1050K to 250K (a 79% reduction) **[3.2, 3.3]**. Also, on the newest Intel CPUs, UTEE supports to execute three orders of magnitude larger dataset sizes, compared to the dataset sizes that are supported to execute by all the competitors **[3.2, 3.3]**. When UTEE and the competitors (e.g., UC Berkeley's Opaque system) run the same dataset sizes supported by the competitors, UTEE achieves over 3.7X performance improvement **[3.2]**.

In short, regarding all TEE-relevant (i.e., aiming to eliminate the OS kernel from the trusted computing base) bigdata computing systems in the world, UTEE could be (one of) the most secure (lowest TCB), most big-data friendly, and highest-performance system.

In terms of impact of UTEE from governments, below is the list of strategic HK-governmental funds, with Prof. Cui as the principal investigator (PI, or leader of invention).
[2.1] Dr. Heming Cui, the project coordinator (project leader recognized by RGC), "MindPipe: High-performance and Carbon-efficient Four-dimensional Parallel Training System for Large AI Models", RGC Research Impact Fund (Ref: R7030-22), HKD $5.4 million, 2023 - 2026.
[2.2] Dr. Heming Cui was awarded **three relevant HK governmental research grants** (Research Grant Councils General Research Funds, or **RGC GRF, all three grants as himself the PI**): (1) "New Systems and Algorithms for Preserving Big-data Privacy in Clouds", GRF (Ref: 17202318), 2019~2022; (2) "Micro-kernel Inspired Systems and Algorithms (MISA): Enabling Secure, Reliable and High-performance Micro-services on Public Clouds", GRF (Ref: 17208223), 2024~2027; (3) "Achieving High-performance and Reliable Transaction/Analytical Processing in Edge Computing", GRF (Ref: 17204424), 2025~2028.
[2.3] HK ITC, ITSP Platform grant award (an award letter is sent to HKU RSS and CS in May 2022, ref: GHP/169/20SZ). PI is Dr. Heming Cui. Grant title is "UTEE: A Secure, Efficient, and Portable Distributed Bigdata Computing System on Heterogeneous Trusted Execution Devices", the same as the title of this KE Award. Grant amount is HK $3.3 million, including 10% of the amount donated from Huawei. Grant period is June 2022 to May 2024.
[2.4] HKU Faculty (Engineering) Knowledge Exchange Award (there is only one award in every year, among the hundreds of faculty members in the Faculty of Engineering of HKU), July 2022, PI: Dr. Heming Cui, award title: "UTEE: A Secure, Efficient, and Portable Distributed Bigdata Computing System on Heterogeneous Trusted Execution Devices" (https://www.ke.hku.hk/assets/doc/KEAward/2022/Engineering.pdf)

3. References to the research (indicative maximum of six references)
Period when the underpinning research was undertaken: 2020 to 2025

HKU PhD/MPhil students, who are primarily supervised by Prof. Cui, are marked with '___'.
Published referred research papers in international top venues on computing systems or security:
[3.1 Cronus ACM Micro 2022] Jianyu Jiang, Qi Ji, Tianxiang Shen, Xusheng Chen, Shixiong Zhao, Sen Wang, Li Chen, Gong Zhang, Xiapu Luo, and **Heming Cui**, CRONUS: Fault-isolated, Secure and High-performance Heterogeneous Computing for Trusted Execution Environments, Proceedings of the 55th ACM/IEEE International Symposium on Microarchitecture. **ACM open source and results reproduced badge**.
[3.2 Uranus ACM AsiaCCS 2020] Jianyu Jiang, Xusheng Chen, Tzs On Li, Cheng Wang, Tianxiang Shen, Shixiong Zhao, **Heming Cui**, Co-Li Wang, and Fengwei Zhang, Uranus: Simple, Efficient SGX Programming and Its Applications, Proceedings of the 15th ACM ASIA Conference on Computer and Communications Security. According to Table 6 of this paper, TCB of LSDS (1050K); TCB of UNC's Civit (1517K); TCB of our Uranus (UTEE) system, 220K.
[3.3 USA patent, about our Uranus (UTEE) system] Jianyu Jiang, Xusheng Chen, Cheng Wang, **Heming Cui**, Sen Wang, Peng Wang, and Gong Zhang, Uranus: An Efficient, Secure Big-data Processing and Programming System based on TEE (USA Application No. 17975163, submitted in Oct 27, 2022).

4. Details of the impact (indicative maximum 750 words)

Compared to our worldwide most advanced and relevant competitors (including Microsoft's VC3, European Union's SGX-Spark, UC Berkeley's Opaque, Google's Confidential Computing, and UNC's Civet), our UTEE system has three notable features below, with explicit evidence based on our six most relevant research references, including **three research papers, two patents, and one grant, all listed in Section 3 (References to the research)** of this document:

[**Smallest trusted computing base (TCB)**] In any security system, TCB means the set of all software and hardware components that must be assumed to be fully trusted and secure of the system; the smaller the TCB of a system, the more secure the system is. Our UTEE system is unique in that it has the smallest TCB among the competitor systems, with four aspects of impact evidence. **First**, claimed in our paper [Uranus AsiaCCS] and our patent [USA patent 1], because UTEE is the world's first work to support 'running only the bigdata queries (including all the decrypted data and sub-functions) within a CPU processor secure execution zone', and the rest of the cloud software (including OS and the big-data computing software which invoke the bigdata queries), are all made by UTEE automatically to run outside the execution zone, UTEE's TCB is around millions of lines of code smaller than all competitor systems. **Second**, explicitly quoted in the 3rd paragraph of **Huawei letter**, Huawei appreciated several novel techniques in UTEE, especially its 'low TCB', and so it is 'suitable to deploy on clouds, but also on personal computers and mobile phones'. **Third**, claimed in our [Cronus ACM Micro 2022] paper, UTEE requires only the CPU processors to contain TEE hardware, and UTEE is the first work to enable software-style-TEE in general GPUs (i.e., these GPUs' hardware do not need to support TEE). These three reasons make UTEE a unique system to support almost all world's general GPU brands, and UTEE being usable by Huawei's 3 million customers in 170 countries (this significant user base is confirmed in the third paragraph of **Huawei letter**). **Fourth**, about the low TCB and security, UTEE passed the China government's most rigor examinations and testing (see Media Report (5) in Section 5).

[**Big-data friendly and Highest performance**] UTEE achieves two aspects of impact. **First**, the concrete evaluation results of our research paper [Uranus AsiaCCS] and our patents [USA patent 1 and 2] contain two facts:: (a) UTEE is world's first system that enables the TEE-enabled secure executions on typical bigdata datasets (e.g., terabyte on some typical bigdata queries and algorithms); (b) this size of dataset being able to run in UTEE (Uranus) is two or three orders of magnitudes larger than those datasets being able to run by competitor systems'. **Second**, praised in the 3rd paragraph of **Huawei letter**, Huawei explicitly praised that, UTEE has the novel technique of 'big-data aware high-performance processing'.

[4.1] Dr. Heming Cui was awarded **three relevant, technology-transfer-oriented research grant from Huawei**, all grants as himself the PI, with a total amount paid from Huawei to HKU (as licensing fee) of HKD $7.5 million: (1) "Architecture, Theory, and Algorithm Research for Accelerating Database Based on Accelerators", Huawei Theory Lab Flagship, HKD $2.55 million, 2023~2025; (2) "ParaNAS: High-performance, Scalable, Reliable and High-precision Multi-GPU Pipeline Parallel DNN Training Systems", Huawei Theory Lab Flagship, HKD $2.4 million, 2021~2023; and (3) "A Blockchain-powered, Trustworthy Internet Layer (System) and its Decentralized and Efficient Applications", Huawei Innovation Research Program (HIRP) Flagship, HKD $2.2 million, 2018~2021.
[4.2] 13 patents licensed to Huawei and applied for patents in China (CN 202411227529.7, CN 202410650032.X, CN 202210138879, CN 202310158460.6, CN 202210138879.0, CN 202111372034.X, 92000692CN01, CN 2021101523346.3, CN 202111080651.2, CN 202010506698.X, CN 202110048599.6, CN 202010366539.4, CN 202010247629.1).

5. Sources to corroborate the impact (indicative maximum of 10 references)
[5.1] Global Times: “Huawei launches six new products to bolster growth in cloud sector” (https://www.globaltimes.cn/page/202104/1222038.shtml).
[5.2] China Daily (中国日报): “华为云可信智能计算服务 TICS，安全释放数据价值” (https://caijing.chinadaily.com.cn/a/202104/29/WS608a7f1ea3101e7ce974cf35.html).
[5.3] China government testing and review report: “12 月 18 日，2020 数据资产大会在北京召开。会上，中国信息通信研究院为通过“基于可信执行环境的数据计算平台”的产品颁发证书。华为云可信智能计算服务 TICS 基于鲲鹏 TrustZone 机密计算，结合硬件 TEE 和软件 SMPC 算法，实现了软硬结合的计算加速，同时支持跨信任域的联邦 SQL 分析和联邦学习能力，以九大测试项全部通过，树立隐私计算产品的新标杆。” (https://icode.best/i/87602637140406).
[5.4] The TelecomReview.com: “Huawei launches 6 new products to power up Cloud business. Trusted Intelligent Computing Service (TICS): HUAWEI CLOUD TICS provides security for innovative and converged applications of data elements, enabling secure data flow and maximizing data value.” (https://www.telecomreview.com/articles/telecom-vendors/4880-huawei-launches-6-new-products-to-power-up-cloud-business).
[5.5] https://huaweicloud.com/intl/zh-cn/news/2020/20210604100805740.html
Consider just one publicly known example about Huawei’s many global strategic company partners: this news reveals that, the China Citic Bank International announced its Huawei Cards in 2021; also, this bank and Huawei jointly announced in this news that, **the Huawei Cards have integrated our UTEE system (TICS) in order to protect the user data security on every credit card transaction**.
[5.6] https://www.cls.cn/detail/1473982
The number of users of this specific bank’s credit card **exceeded 2 millions in Sep 25, 2023**.
[5.7] https://post.smzdm.com/p/a5xgg378/
The number of applicants of the Huawei Cards among all Huawei’s company partners **exceeded 10 millions in Sep 2, 2024** (no source to justify that other partners also use TICS; but for two reasons: (1) the purpose of saving development team effort and money for Huawei, likely other partners also use the same software implementations, including UTEE, in all Huawei Cards; (2) according to public news, Huawei has not released other secure clous service other than our UTEE TICS, and Huawei should not provide a credit card service to other companies with the same type/name of credit card but with weakened security guarantees (i.e., removing UTEE for other partners’ credit card implementations).
[5.8] https://hemingcui.github.io/doc/utee-ack.pdf
In 2021, Huawei’s formal acknowledgement letter about the commercialization of UTEE (also attached in the next page of this form) reveals that, UTEE has 3 million active cloud users (including independent companies and individuals) located in 170 countries.

Letter of Appreciation for Contribution to Technical Cooperation Project

To whom it may concern,

I, on behalf of the Theory Lab, Huawei Technologies Co., Ltd, am writing to show our commercialization acknowledgement of UTEE (Ubiquitous Trusted Execution Environments): a secure, efficient, and portable distributed big-data computing system based on advanced CPU secure hardware components (e.g., ARM TrustZone and Intel SGX). UTEE is a novel eco-system combining the complementary security advantages of these hardware components and high-level programming languages (e.g., Java and Scala). UTEE can effectively prevent the computations and queries on plaintext data being peeked by diverse malicious parties on public clouds, including operating systems, hypervisors, and malicious/careless cloud administrators.

My research team in the Huawei Theory Lab has been looking for solutions of secure big-data computing systems built by world-wide leading universities and industries. In 2018, we offered a two-year flagship research fund of HK $2,189,450 to Dr. Heming Cui on jointly developing such systems. From this research fund, we have conducted substantial collaborations with Dr. Cui and produced diverse research achievements, including jointly publishing a series of research papers in international top journals/conferences (e.g., TDSC, TPDS, EuroSys, DSN, SRDS, and Asia CCS) and applying for seven patents. Below are three selected papers and two selected patents relevant to secure big-data computing systems. UTEE is most relevant to the last patent.

- Uranus: Simple, Efficient SGX Programming and Its Applications, 15th ACM ASIA Conference on Computer and Communications Security (ASIACCS 2020).
- DAENet: Making Strong Anonymity Scale in a Fully Decentralized Network, IEEE Transactions on Dependable and Secure Computing (TDSC 2021).
- UPA: An Automated, Accurate and Efficient Differentially Private Big-data Mining System, 50th IEEE/IFIP International Conference on Dependable Systems and Networks (DSN 2020).
- An Automated, Accurate and Efficient Differentially Private Big-data Mining System, submitted to CNIPA, Ref. CN 202010506698.X.
- An Efficient, Secure Big-data Processing and Programming System based on Trusted Execution Environment, submitted to CNIPA, Ref. CN 202010366539.4.

UTEE is one major resultant system developed from our collaborations. Compared to UTEE's worldwide competitors (e.g., Alibaba, Baidu and Microsoft), we appreciate several unique novel techniques in UTEE, including its low trusted computing base, big-data aware high-performance processing, and portability across heterogenous CPU vendors. These unique techniques of UTEE make it not only be suitable to deploy on clouds, but also on personal computers and mobile phones. In April 2021, our Huawei Clouds CEO announced the release of Trusted and Intelligent Cloud Services (TICS, see https://www.huaweicloud.com/product/tics.html), and UTEE has been successfully deployed in TICS as a core system component. This means that UTEE is usable by the 3 million Huawei Clouds customers (enterprises and individuals) located in 170 countries.

Overall, we believe the recognition of Dr. Cui's achievements on your side will be a major encouragement of his future research on developing secure systems that are both academically innovative and commercially practical. These developments can contribute to the effective protection of data for many entities, including individuals, enterprises, organizations, and governments.

Bo Bai

Sincerely,

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